

SHADMAN MUHTASIM

Software Engineer • Backend & Full-Stack Developer

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PROFESSIONAL SUMMARY

Computer Science undergraduate specializing in backend and full-stack software development with a strong **ASP.NET Core (.NET 8)** foundation and practical **AI/LLM application engineering** exposure. Proficient in architecting secure RESTful APIs with **JWT authentication**, role-based access control (**RBAC**), and layered architectures, alongside integrating LLM APIs and tool-calling conversational agents into practical workflows. Proven track record of authoring automated test suites, containerizing multi-tier services with **Docker**, and collaborating on academic and hackathon engineering projects.

EDUCATION

Ahsanullah University of Science and Technology (AUST)

Dhaka, Bangladesh

Bachelor of Science in Computer Science and Engineering

Expected Graduation: 2027

Current Standing: 6th Semester (2 Semesters Remaining)

CGPA: **3.135 / 4.00**

TECHNICAL SKILLS

Languages: C#, TypeScript, JavaScript, SQL, Python, C, C++, PHP

Backend & Frameworks: ASP.NET Core (.NET 8), EF Core, ASP.NET Core Identity, RESTful APIs, Laravel

Frontend: React, Vite, TypeScript, Tailwind CSS, TanStack Query, shadcn/ui

Databases & ORM: SQL Server, MySQL, SQLite, PostgreSQL, Prisma ORM

AI & LLM Engineering: Gemini API, Native Function/Tool Calling, Multi-Provider LLM Integration, Prompt Engineering

DevOps & Testing: Docker, Docker Compose, Git, GitHub Actions, xUnit, Moq, Vitest, Postman, Linux

PROJECTS

Karigor — Service Marketplace Platform

Full-Stack Project

Technologies: C#, ASP.NET Core, EF Core, SQL Server, React, TypeScript, TanStack Query, Tailwind CSS

- Architected **ASP.NET Core Web API** backend with **EF Core** and **SQL Server**, enforcing **ASP.NET Core Identity** with role-based access control (**RBAC**: Customer, Worker, Admin) and **JWT authentication** with refresh token rotation.
- Built complete service booking lifecycle, customer request flows, worker onboarding (skills and document uploads), and administrative dashboards.
- Developed responsive **React/TypeScript** frontend with **TanStack Query** for asynchronous server-state cache invalidation and UI state synchronization.

LearnifyAI — AI-Powered Learning Platform

Backend & AI Integration

Technologies: C#, .NET 8, ASP.NET Core, EF Core, SQL Server, React, TypeScript, xUnit, Moq

- Designed a layered **.NET 8 architecture** separating API controllers, business services, and **EF Core** data repositories for clean separation of concerns and maintainability.
- Integrated **multi-provider LLM APIs** to generate dynamic educational content, study summaries, and interactive quizzes with background processing services.
- Secured endpoints with **JWT/refresh tokens**, applied **FluentValidation** pipeline rules, and authored automated unit test suites using **xUnit** and **Moq**.

CarriGrow — Job Portal with Mentorship Platform

Backend Developer • Academic Team Project

Technologies: Laravel (PHP), MySQL, REST API, Docker, Docker Compose, React

- Implemented backend **RESTful APIs** using **Laravel (PHP)** for job search workflows, applicant tracking, and mentorship scheduling.
- Designed normalized **MySQL** database schemas with foreign-key integrity and optimized queries for user roles, mentor profiles, and job postings.
- Containerized services with **Docker** and **Docker Compose** to achieve consistent local development and testing environments.

CampusOS — AI-Powered Campus Management Platform

CSE Carnival 8 Hackathon (+6.1k Lines)

Technologies: Next.js 14, TypeScript, Prisma ORM, SQLite, Google Gemini 2.0 Flash, Vitest, Docker

- Implemented full-stack **CRUD APIs**, server service layers, and administrative domain dashboards across 5 campus data modules (+6,100 authored lines).
- Engineered reliable conversational AI workflows by debugging agent endpoints, implementing **Gemini 2.0 Flash** native tool/function calling with typed schemas, and configuring model fallback handling.
- Enforced strict schema validation using **Zod** and **Prisma ORM**; authored automated **Vitest** suites to validate business logic.

RELEVANT COURSEWORK & LANGUAGES

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Software Development, Database Systems, Computer Networks, Operating Systems, Artificial Intelligence, Information System Design & Software Engineering

Languages: Bangla (Native), English (Professional Working Proficiency), Hindi (Professional Working Proficiency)